

Politecnico di Torino

Politecnico di Torino

Master of science program in CYBERSECURITY

A.a. 2025/2026

Graduation Session March 2026

An Experimental Framework for Evaluating Secure Deletion Techniques in Modern Digital Environments

Relatori:

Andrea Atzeni

Giuseppe Emiliano Vaciago

Candidati:

Imesha Ann Anjalie Ponnamperumage

Abstract

Acknowledgements

Table of Contents

List of Tables v

List of Figures vi

1 Introduction 1

2 Background and Related Work 3
- 2.1 A Taxonomy of Anti-Forensic Techniques 4
- 2.2 Secure Deletion; Challenges and Limitations 5
- 2.3 Traditional Secure Deletion Techniques 6
- 2.4 The Flash Memory Challenge 6
- 2.5 Secure Deletion in Virtualized and Containerized Environments . . 7
- 2.6 Legal and Ethical Dimensions 7
- 2.7 Summary and Identified Research Gaps 7

3 Systematic Review of Secure Deletion Methods 8

4 Experimental Framework and Methodology 9

5 Experimental Results and Analysis 10

6 Interpretation of Findings and Implications 11

7 Conclusions and Future Work 12

A Appendix A 13

Bibliography 14

List of Tables

List of Figures

Chapter 1

Introduction

Secure data deletion — the task of rendering data irrecoverable from physical storage media — has become a critical concern across cybersecurity, regulatory compliance, and digital forensics. In the physical world, the importance of destroying sensitive information is well understood: confidential documents are shredded, and storage media containing classified material are physically destroyed. In the digital domain, the problem is significantly more complex. Standard file deletion operations, implemented across all modern file systems, simply unlink the file's metadata entry while leaving the full contents intact on the storage medium [1]. Users typically assume, falsely, that once they delete a file it is irrecoverable.

The urgency of this problem has been compounded by two converging forces. First, regulatory frameworks such as the GDPR right to erasure, NIST SP 800-88 Rev. 1, and IEEE 2883-2022 now mandate demonstrable, verifiable data removal at defined sanitization levels — Clear, Purge, and Destruct — each specifying the required assurance against data recovery [2]. Second, the widespread transition from Hard Disk Drives (HDDs) to Solid-State Drives (SSDs) has fundamentally altered the storage landscape, rendering decades of established secure deletion practice inadequate.

Unlike HDDs, SSDs employ a Flash Translation Layer (FTL) that maps logical block addresses to physical flash pages. Due to the erase-before-write property of NAND flash memory, the FTL performs out-of-place updates: modified data is written to a new physical location while the old page, still containing the original data, is marked as invalid. Wear-leveling algorithms further disperse data across the device to distribute program-erase cycles, and background garbage collection reclaims invalid pages only when the pool of free blocks runs low. Over-provisioning reserves additional memory invisible to the host operating system, where residual copies of deleted data may persist indefinitely. These mechanisms collectively ensure that a conventional delete operation at the operating system level does not reliably remove the underlying data, creating a persistent risk of data remanence

[1, 2].

Chapter 2

Background and Related Work

The growing dependence on digital systems for both personal and professional activities has resulted in an unprecedented accumulation of sensitive information. As organizations and individuals increasingly store confidential data on a wide range of digital media—ranging from traditional hard drives to solid-state drives and cloud environments—the need for reliable and verifiable data deletion mechanisms has become critical. Failure to ensure proper data sanitization can expose users to privacy breaches, corporate espionage, and regulatory non-compliance.

From a cybersecurity perspective, secure deletion plays a dual role. On one hand, it is an essential component of data protection and privacy preservation, ensuring that information is irretrievably removed when no longer needed. On the other hand, it can also be employed as part of anti-forensic strategies, deliberately obstructing digital investigations by erasing potential evidence. Understanding this duality is fundamental to appreciating the technical and ethical challenges that surround secure deletion.

The following sections provide an overview of the conceptual and technical foundations that inform this research. First, a taxonomy of anti-forensic techniques is presented to situate secure deletion within the broader context of digital obfuscation strategies. Then, the principles of data deletion and persistence are examined, followed by an analysis of traditional and modern deletion approaches across different storage technologies and virtualized environments. Finally, the legal and ethical implications of secure deletion are discussed, highlighting the complex balance between privacy rights and forensic accountability.

This background establishes the theoretical basis for the experimental framework proposed in this thesis, which aims to evaluate and compare the effectiveness of secure deletion methods across contemporary computing infrastructures.

2.1 A Taxonomy of Anti-Forensic Techniques

The field of digital forensics is founded on the assumption that traces of user activity persist within digital systems and can be extracted for analysis and attribution. Anti-Forensics challenges this assumption by allowing the deliberate use of various techniques, procedures, tools and stratergies to obstruct the collection of digital evidences from a computer system. The main objective of these methodologies are to hinder Data Collection and impede Forensic investigations. These techniques aim to obscure, alter, or destroy digital evidence in ways that reduce the likelihood of recovery or interpretation. González Arias, et al. [3] provide one of the most comprehensive systematic reviews of this domain, analyzing 220 studies spanning from 2006 to 2022. Their taxonomy identifies four overarching categories of anti-forensic activity: **data destruction, artefact deletion, obfuscation, and attacks on forensic tools and methods.**

- **Data destruction** refers to the deliberate and irreversible removal of information, often achieved through overwriting, encryption, or physical destruction.
- **Artifact deletion** refers to the deliberate deletion of data to remove traces of activity—this is where *secure deletion* techniques become highly relevant.
- **Obfuscation** aims to mislead investigators disguising the nature, structure, or location of data to hinder recognition and analysis (e.g., steganography or data fragmentation).
- **Attacks of Forensic Tools and Methods** exploit vulnerabilities in forensic software or hardware to corrupt or mislead analysis.

This review highlights that while forensic techniques continue to evolve, so too do the methods employed to counteract them. But anti-forensic strategies remains a neglected research field, lacking standardized terminology, empirical testing, and benchmark datasets which makes it difficult to compare results across studies or build systematic defenses. Understanding these categories provides context for why secure deletion is often used for privacy or compliance, and also as an anti-forensic tool. However, this dual use highlights the ethical balance between data protection and evidence preservation that this research seeks to explore. Secure deletion has an unclear place in this classification. If it is used to hide criminal conduct, it could be considered a purposeful anti-forensic act, although it could fulfill lawful privacy or regulatory compliance aims, such as the EU's *right to erasure.* Context and intent make the difference, as Arias et al note, [3] "the same technique that safeguards confidentiality can, in adversarial hands, obstruct justice." Therefore, any balanced cybersecurity framework must take both forensic countermeasures and privacy-preserving computing into account when studying secure deletion.

2.2 Secure Deletion; Challenges and Limitations

Data deletion in digital systems is far more complex than its terminology suggests. Deleting a file from an operating system interface rarely removes its physical presence from the storage medium. Instead, the system merely updates metadata; marking sectors as available for reuse while leaving the original data intact until overwritten. Joukov, Papaxenopoulos, and Zadok [4] famously identified several pervasive misconceptions about secure deletion that persist even among professionals. Chief among these is the assumption that encryption alone guarantees data destruction, or that multiple overwrites are necessary for secure erasure. Their research demonstrated that a single, atomic overwrite operation is generally sufficient for magnetic storage, provided it is executed consistently and covers all data replicas, including journaling and metadata. However, the authors also highlighted the challenge of maintaining crash consistency — ensuring that deletion operations complete even in the event of system interruptions or power loss. This observation remains crucial in the context of modern SSDs, where caches and mapping tables may retain data even after deletion is logically completed.

Reardon, Basin, and Čapkun [1] contributed a foundational Systematization of Knowledge (SoK) on secure data deletion, bridging the gap between disparate methods and definitions. They proposed a layered taxonomy consisting of the user layer, file system, device driver, controller, and physical hardware. Each layer offers varying trade-offs between identifiability (the precision with which data can be targeted for deletion) and irrecoverability (the assurance that deleted data cannot be restored).

Their analysis revealed that higher layers (e.g., application or file system) possess semantic understanding of what to delete but lack physical control over the underlying media. Conversely, lower layers (e.g., hardware controllers) can permanently erase data but cannot discern its logical context. The tension between these layers defines the central challenge of secure deletion: achieving both semantic correctness and physical assurance.

Moreover, Reardon et al. [1] emphasized the importance of considering adversary models — differentiating between threats such as casual software recovery, forensic examination, or physical chip-level analysis. A deletion method effective against software recovery may still be vulnerable to forensic or hardware attacks. Thus, any secure deletion solution must be context-aware, calibrated to the adversary's capabilities and the sensitivity of the stored data.

2.3 Traditional Secure Deletion Techniques

Historically, secure deletion relied on *overwriting* — writing random or fixed patterns over the data to prevent recovery. Tools like shred, srm, and dd were widely used for this purpose. Standards such as the **U.S. Department of Defense (DoD 5220.22-M)** specified multiple overwrite passes to ensure data erasure on magnetic media.

However, as storage technologies evolved, it became clear that overwriting is not always reliable:

- Logical and physical mapping layers (e.g., in SSDs) can redirect writes, leaving original data intact.
- Wear leveling mechanisms in flash memory complicate direct access to physical storage locations.
- Modern drives may maintain hidden or reserved areas inaccessible to user-level commands.

While traditional techniques remain relevant for certain media, they no longer guarantee complete data sanitization on modern hardware.

2.4 The Flash Memory Challenge

Flash-based storage, including SSDs and USB drives, introduces unique challenges for secure deletion due to its internal **wear-leveling algorithms and block-based architecture**. Unlike magnetic disks, flash memory cannot overwrite data directly; instead, it must erase entire blocks before writing new data. This results in copies of deleted information potentially persisting in unaddressed cells.

Additional complications arise from:

- **Over-provisioning:** Manufacturers reserve hidden blocks for performance, inaccessible to standard overwriting commands.
- **Garbage collection:** Automated internal processes may relocate or temporarily preserve deleted data.
- **TRIM command behavior:** While intended to improve SSD performance, TRIM's effect on secure deletion is inconsistent across devices and operating systems.

These characteristics make flash memory a critical focus area for experimental evaluation frameworks like the one proposed in this thesis.

2.5 Secure Deletion in Virtualized and Containerized Environments

While secure deletion has been extensively studied in classical filesystems, flash storage, cloud storage, and virtual machines, there is virtually no focused research evaluating secure deletion inside containerized environments (e.g., Docker, containerd, Kubernetes ephemeral storage). This absence highlights the need for foundational work in defining threat models, measuring residual artifacts, and establishing deletion guarantees in layered container storage systems.

2.6 Legal and Ethical Dimensions

2.7 Summary and Identified Research Gaps

Chapter 3

Systematic Review of Secure Deletion Methods

Chapter 4

Experimental Framework and Methodology

Chapter 5

Experimental Results and Analysis

Chapter 6

Interpretation of Findings and Implications

Chapter 7

Conclusions and Future Work

Appendix A

Appendix A

```
1     const test = test;
```

$\mathcal{O}\left(n \log n\right)$
`numpy`

Bibliography

[1] Joel Reardon, David Basin, and Srdjan Čapkun. «SoK: Secure Data Deletion». In: *Proceedings of the 2013 IEEE Symposium on Security and Privacy (SP'13)*. San Francisco, CA, USA: IEEE, 2013, pp. 301–315. DOI: 10.1109/SP.2013.28 (cit. on pp. 1, 2, 5).

[2] Fabian Aronsson and Oskar Lund. «Secure Data Deletion: Ensuring Confidentiality in Digital Systems ; A Survey of Methods for Data Erasure». LIU-IDA/LITH-EX-A–2025/074–SE. Master's thesis. Linköping, Sweden: Linköping University, 2025. URL: http://www.ep.liu.se/ (cit. on pp. 1, 2).

[3] Rafael González Arias, Javier Bermejo Higuera, J. Javier Rainer Granados, Juan Ramón Bermejo Higuera, and Juan Antonio Sicilia Montalvo. «Systematic Review: Anti-Forensic Computer Techniques». In: *Applied Sciences* 14.12 (2024). ISSN: 2076-3417. DOI: 10.3390/app14125302. URL: https://www.mdpi.com/2076-3417/14/12/5302 (cit. on p. 4).

[4] Nikolai Joukov, Harry Papaxenopoulos, and Erez Zadok. «Secure deletion myths, issues, and solutions». In: *Proceedings of the Second ACM Workshop on Storage Security and Survivability*. StorageSS '06. Alexandria, Virginia, USA: Association for Computing Machinery, 2006, pp. 61–66. ISBN: 1595935525. DOI: 10.1145/1179559.1179571. URL: https://doi.org/10.1145/1179559.1179571 (cit. on p. 5).